F78AB - Actuarial and Financial Mathematics B

Computer Lab 1 – Index-linked bonds

Suppose that today is 22 January 2013.

The last published RPI figure is RPI(Dec 2012)= 246.8. The RPI for November 2012 was 245.6. Assume that RPI will increase in the future at a constant effective rate of 2.5% per annum.

Today's market price of £100 nominal of the $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 is £323.57, just after the payment of the January 2013 coupon. The $4\frac{1}{8}\%$ Index-linked Treasury Stock 2030 was issued in June 1992 and has an 8-month time lag. Its Base RPI figure is 135.1, the October 1991 RPI.

Coupon dates: 22 January, 22 July. Redemption date: 22 July 2030.

Question 1 Using a calculation date of 22 January 2013, create a spreadsheet that shows the projected money payments at each payment date. Then use Excel to find the money yield.

In particular, your spreadsheet should show:

- the payment dates. The Excel training 1.1 weblink in VISION, under *Learning Materials* → *Computer Labs* → *Excel Training*, tells you how to increase each date by 6 months using the Excel command *DATE*.
- the unindexed cashflows occurring at each payment date. Give the unindexed annual coupon amount an Excel *name*, and give the redemption amount an Excel *name*. Use the two names, and not the values, when typing in the unindexed cashflows. The Excel training 1.2 weblink in VISION tells you how to create Excel names.
- the 8-month lagged RPI dates. Again, this can be calculated automatically by using the *DATE* command to subtract 8 months from the payment dates.

Best practice

- Your calculations to determine the indexed cashflows should be broken down in small steps. This is best practice as it makes your calculations easier to check and follow.
- As far as possible, you should assign a name to each constant in Excel. For example, the time lag, the time between indexation dates, the base RPI figure, the constant inflation rate, and so on, should all be assigned a name. This is best practice as it makes your spreadsheet easier to modify if these constants change or if you want to adapt the spreadsheet to evaluate another bond.

Once you have found the estimated future money cashflows of the bond,

- choose an arbitrary interest rate and assign it a name.
- discount the indexed cashflows to the calculation date 22 January 2013.
- use *Goal Seek* (under *Data* → *What-If Analysis*) to find the interest rate that equates the price of the bond calculated in your spreadsheet to the market price of the bond £323.57 per £100 nominal. The Excel training 1.3 weblink in VISION tells you how to use *Goal seek*.

What happens to the price of the bond calculated in your spreadsheet when the constant inflation rate is

(a) 2% per annum? (b) $3\frac{1}{8}\%$ per annum?

Can you explain why?